

## SIGNLOOP APP SUMMARY

### WHAT IT IS

SignLoop is a Next.js web app for AI-assisted contract review. Users upload contracts, extract text, run structured risk analysis, and review findings.

### WHO IT IS FOR

- Primary user: people reviewing business contracts and triaging risk before signing.
- Exact persona/job title: Not found in repo.

### WHAT IT DOES

- Provides an authenticated workspace (Clerk sign-in, dashboard, settings).
- Uploads contracts (PDF, image, txt, doc, docx) with server-side MIME validation.
- Extracts text using unpdf/Word parsing, with OCR support via Tesseract for images.
- Runs AI analysis and stores structured output (risk badge, key points, red flags, key dates, next actions).
- Organizes work by projects with linked contracts and context documents.
- Persists data in Postgres and stores files in Vercel Blob (local filesystem fallback in dev).
- Allows primary model selection; analysis pipeline includes JSON normalization/validation and provider fallback.

### HOW IT WORKS (ARCHITECTURE)

- UI: Next.js App Router pages in apps/web/app with React Query data fetching.
- Auth: Clerk middleware protects non-public routes and API endpoints.
- API: Next.js route handlers under apps/web/app/api for contracts, projects, uploads, analysis, jobs, and settings.
- Processing: lib/text-extraction.ts handles PDF/image/text/Word extraction; lib/analysis.ts calls LLM providers and validates JSON with Zod schemas.
- Data: lib/server-db.ts manages Postgres tables (projects, contracts, analyses, context\_documents, contract\_files, user\_settings).
- Storage: lib/object-storage.ts writes to Vercel Blob when token exists, otherwise local disk.
- Main flow: create contract -> upload file -> extract/save text -> analyze -> persist analysis -> render in /contracts/[id].
- Current limitation: project context documents are stored/listed, but analysis currently runs on contract text only.

### HOW TO RUN (MINIMAL)

- From repo root: npm install
- Configure required environment variables listed in README.md
- Start the workspace: npm run dev
- Open the web app: http://localhost:3000

### MISSING SETUP ARTIFACTS

- Example env template (.env.example or similar): Not found in repo.